


## MATHEMATICAL FORMULATIONS, RISK BOUNDS & REAL-TIME SYSTEM PERFORMANCE SLAS

**Tri-Band Temporal Decay:** $\Phi_{\text{time}}(\Delta t) = [\cos(\omega_k \Delta t), \sin(\omega_k \Delta t)]_{k=1}^{3}$

Adaptive Edge Trust: $\hat{g}_{ij} = \sigma(\mathbf{w}_g^{\top}[\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}]), \quad \mathbf{A}_{\text{pruned}} = \mathbf{A} \odot \mathbf{I}(\hat{g}_{ij} \geq \delta_{\text{floor}})$

12-D Flow Invariants: $\Phi_{\text{flow}} = \ln(1 + \sum A_{\text{out}}/(\sum A_{\text{in}} + \epsilon)), \quad \lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha e^{-\beta(t - t_i)}$

**Class-Conditional CRC:** $\hat{q}^{(y)} = \text{Quantile}_{1-\alpha_y}(\{s_i^{(y)}\}_{i \in \mathcal{D}_{\text{cal}}^{(y)}}), \quad \mathbf{P}(y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha$

Triage Dispatch: $\Gamma(u) = \{y \in \{0,1\} : s_u^{(y)} \leq \hat{q}^{(y)}\}, \quad \mathbf{E}[|\Gamma(X)|] = 1.0050, \quad \mathbf{P}(\Gamma = \varnothing) = 0.00\%$

**Fast-Path Latency SLA: Single-event** $0.45\,\text{ms} < 10\,\text{ms}$ **SLA | Review Queue:** $-99.50\%$ **Workload Cut | Peak VRAM:** $428\,\text{MB}$